Information

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Team F:

Teammates: Leena Aljurashi, Thomas Wang, Dillon Ma

ILR02

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Individual Progress

Since the previous lab, I worked on researching on motors alongside the calculations needed to figure out weights. While searching, I also looked more into 3d printed gearboxes for the possibility that we would need tot gearbox the motors. In the end, I would decide that we would use 6 powerful DC motors for the 6 “yaw” parts, 1 for each leg. This is due to each part needing to push and pull the most weight. The other 12 joints, or 2 per leg, will be driven by servos. For the 6 motors, I am currently looking to use a motor with either a built-in encoder, or an external encoder to use alongside the motor. Through my calculations, each motor will need to be able to have a torque force of atleast 30-37.5kg.

Also since the previous lab, I started to work on the Power system diagram. Throwing an initial sketch, I would help design where each thing would go.

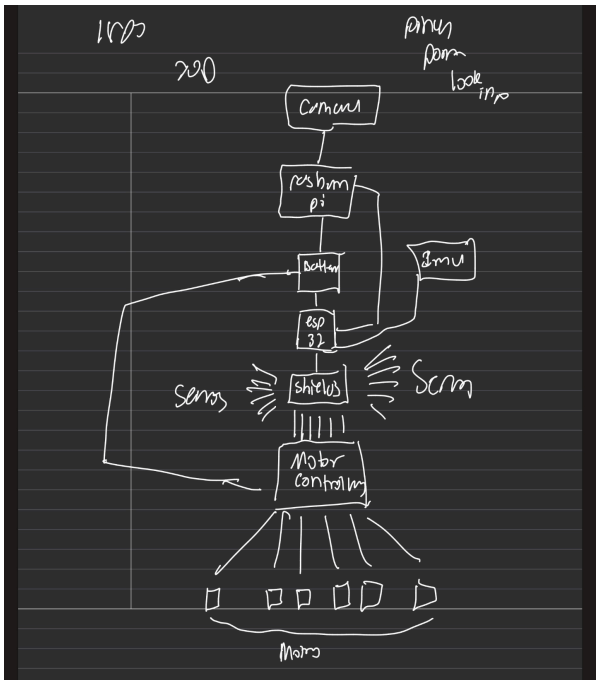

Figure 1: Sketch of the power system diagram

Challenges

I would have issues trying to find a balance with motors and fitting the budget with still having 12 servos that are beefy enough. I would collect a list of motors to compare, and while working on said list, I would encounter an issue with my document, where it would wipe some of the links I had saved. Currently, I am working on recovering the current links.

Furthermore, for the power system diagram, after getting feedback from Professor Travers, alongside the TA’s, the storage of the battery alongside the issue of one singular battery powering everything came up.

Teamwork

While I was working on my parts, each member was working on their respective system. Each person worked on the gnatt chart, as Dillon worked on the general CAD design, and Thomas and Leena worked on the pseudo code for control architecture.

Plans

My future plan is to double check my force calculations with other members of CMU's Robotics Club, then finalize the motor or motor and motor controller, alongside the servos. After this, I plan to work with Dillon to help figure out the final size measurements for each joint and part of the robot. Furthermore, I plan to look more into the power system of the robot, looking into potentially a multiple battery system, all connected to one common ground.

2. 1 to 2 pages of 12-point text (Arial, Calibri, Cambria, Helvetica, etc.: standard fonts, sans serif preferred) 1.15 spacing, 1” margins. The 1-2 pages do not include figure(s).
3. Use a separate heading for four of the five required elements described below (Individual Progress, Challenges, Teamwork, and Plans). The figure(s) should be placed in the text where it makes most sense.
4. PDF format

Content

1. General

a. Text

i. Spend text on explaining and clarifying work and adding depth

ii. Don’t add “filler” material

iii. Proofread, including a spell and grammar check; don’t use slang

iv. Section headings help the reader

b. Gripes (e.g., about teammates, the course in general, etc.) should come in person to us, not in the ILR

c. Do not include raw code in your ILR unless you are explicitly asked to, as is the case for the Sensors and Motor Control Lab. If you want to depict software, see 2.d.ii below.

2. The five required elements

a. Individual progress

i. What did you achieve since the last ILR/presentation? The description should give relevant and specific technical details, not just a brief mention of what you did.

b. Challenges

i. What difficulties or interesting problems did you face since the last ILR/presentation?

ii. Talk about expected challenges in the future under Plans below, not in this section.

iii. A challenge should be accompanied by some level of discussion, for example, an attempt to analyze the source of a problem, or suggested ways for addressing it. It isn’t sufficient to say simply, “we had a problem with A” and nothing more.

c. Teamwork

i. Describe complementary and joint work by teammates, mentioning each of them by name.

ii. The goal of this element is to demonstrate your awareness of how individual work relates to that of other teammates and to the system goals.

d. Figure(s)

i. Various figure types may be used (hand-drawn, CAD, photo (including from a

cellphone), table, plot, flowchart, block diagram, schematic, circuit diagram, etc.).

ii. Do not include a software listing as a figure. If you want to use software as a figure, include a flowchart, pseudocode, or some other abstract representation of the ideas and processes underlying your software.

iii. An ILR may have an unlimited number of figures, but at least one must be unique. A figure is unique if no one else on the team has included it in his ILR. Students also may not recycle their own figures from earlier ILRs.

iv. Non-original figures and non-unique figures should be properly attributed. The former should have some kind of citation, URL, or other reference, and the latter should cite the teammate responsible for the figure.

v. Each figure should have a number, a caption, and possibly labels.

1. The text of the ILR should refer to each figure (by number) in a way which aids the reader's overall understanding. Figures should not be inserted without being referred to in the text.

2. The "caption" is the text immediately after the figure number under the figure. The caption should be below the figure. A "label" is a text box with an arrow that names and points to a specific element in the figure.

3. Individual components in figures should be annotated using labels with pointers or arrows, especially for otherwise complex or confusing figures. Examples are a figure with a) many elements in it whose identity is not obvious without assistance; b) background clutter that requires "calling out" of the foreground elements of interest.

4. Two good examples of numbered, captioned, and labeled figures are shown below (Figures 1 & 2).

Figure 1. Zigbee communication demonstration setup

Figure 2. Robot chassis (digging robot critical subsystem)

e. Plans

i. This should include and focus on work the individual, not just the team in general, intends to do from now until the next lab demo.